

STUDENT PROJECT TEMPLATE (SRS)

(INSTRUCTION FOR STUDENTS: THIS DOCUMENT SERVES AS THE FORMAL 'CONTRACT' FOR YOUR TEAM PROJECT. IT MUST BE MAINTAINED IN YOUR GITHUB REPOSITORY.)

POLY SERVICE

SOFTWARE REQUIREMENTS SPECIFICATION

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TERM: SPRING 2026

SECTION: 09

THE ENGINEERING TEAM:

1. **STUDENT A:** SEAN — SCRUM MASTER
2. **STUDENT B:** EDGAR — PRODUCT TESTER
3. **STUDENT C:** SCOTT — PRODUCT OWNER
4. **STUDENT D:** MARC — LEAD DEVELOPER

PROJECT ASSETS:

- **GITHUB:** [HTTPS://GITHUB.COM/SEAN-LEBLANC14/SERVICE-MARKETPLACE](https://github.com/SEAN-LEBLANC14/SERVICE-MARKETPLACE)
- **DEPLOYMENT:** [HTTPS://SERVICE-MARKETPLACE-SILK.VERCEL.APP/](https://SERVICE-MARKETPLACE-SILK.VERCEL.APP/)

DOCUMENT HISTORY:

LAST DATE CHANGED:	WHO:	WHAT WAS CHANGED:
4/17/2026	SCOTT, MARC, EDGAR, SEAN	REPO LINK, ROLES, ALL STARTING INFO
5/4	SCOTT, MARC, EDGAR, SEAN	FINALIZED PROJECT VISION AND USER STORIES
6/4	SCOTT, MARC, EDGAR, SEAN	FINALIZED THE NON-FUNCTIONAL REQUIREMENTS

SERVICES MARKETPLACE

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

COURSE: INTRODUCTION TO SOFTWARE ENGINEERING / CSC307

DATE: 6/5/2026

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1. INTRODUCTION

- **PROBLEM STATEMENT: WHAT SPECIFIC PROBLEM ARE YOU SOLVING:**

- Cal Poly students often struggle to find reliable and trustworthy local services within their campus community. Existing platforms like Craigslist or Fiverr are not tailored to students and may lack trust, safety, or relevance for campus-specific needs. As a result, students face difficulty connecting with peers for everyday services such as tutoring, moving help, or small jobs.

- **TARGET AUDIENCE: WHO ARE THE STAKEHOLDERS?:**

- CAL POLY STUDENTS

- **SCOPE: WHAT IS THE SCOPE OF THIS PROJECT?:**

- SERVICE EXCHANGE THROUGHOUT SAN LUIS OBISPO

- **PRODUCT VISION:**

- For Cal Poly students who want a safe and convenient way to exchange services, PolyServe is a student-only services marketplace that allows users to offer and purchase services within the campus community. Unlike platforms such as Craigslist or Fiverr, our product is exclusively for verified university students, ensuring a more trustworthy and localized experience.

2. USER STORIES

USER STORIES FOLLOW THE FORMAT: "AS A [TYPE OF USER], I WANT TO [ACTION] SO THAT [VALUE/BENEFIT]."...

- **US-01: AUTHENTICATION**
- **US-02: STORE WORKOUT LOG.**
- **US-03: SEARCH HISTORY**

ID	REQUIREMENT	PRIORITY
US-01 SEAN	As a user, I want to create an account and log in securely so that I can safely access the platform.	1
US-02 SEAN	As a student, I want to register with a university	2

	email so that only verified college students can use the platform.	
US-03 SEAN	As a user, I want to create a profile with personal information so that I can showcase my services and identity.	3
US-04 EDGAR	As a service provider, I want to create a listing with a title, description, and price so that I can offer my services to others.	4
EDGAR	As a user, I want to browse available services so that I can discover services I need.	5
EDGAR	As a user, I want to filter listings by category so that I can narrow down results.	6
SCOTT	As a user, I want to search listings using keywords so that I can quickly find relevant services.	7
SCOTT	As a service provider, I want to edit or remove my listings so that I can keep my offerings accurate and	8

	up to date.	
SCOTT	As a user, I want to pay for a service when I book it so that I can complete transactions easily.	9
MARC	As a service provider, I want to receive payment for services I provide so that I am compensated	10
MARC	As a service provider, I want to view a history of services I have provided so that I can track my work.	11
MARC	As a customer, I want to receive a refund if a service is cancelled so that I can protect myself from losing money.	12
MARC	As a customer, I want to leave ratings and written reviews so that I can share feedback about services.	13
EDGAR	As a service provider, I want to view ratings left by customers so that I can understand my performance.	14
SCOTT	As a customer, I want to see a provider's average rating so that I can choose trustworthy services.	15

SEAN	As a user, I want to receive confirmation when I book and pay for a service so that I know the transaction was successful.	16
SCOTT	As a service provider, I want to receive notifications when someone books and pays so that I can prepare for the service.	17
SEAN	As a service provider, I want to receive notifications when a booking is cancelled so that I can adjust my schedule.	18
EDGAR	As a user, I want to report a service provider for violating guidelines so that the platform remains safe.	19
MARC	As an admin, I want to remove listings or suspend users so that I can enforce platform rules.	20

3. **FUNCTIONAL REQUIREMENTS**

THE SYSTEM SHALL...

ID	REQUIREMENT	PRIORITY
FR-01	The system shall allow users to register and authenticate securely.	1
FR-02	The system shall allow users to create, edit, and delete service listings.	2
FR-03	The system shall allow users to search and filter listings.	3
FR-04	The system shall allow users to book and pay for services.	4
FR-05	The system shall allow users to leave ratings and reviews.	5
FR-06	The system shall send notifications for bookings and cancellations.	6
FR-07	The system shall allow admins to moderate listings and users.	7

4. NON-FUNCTIONAL REQUIREMENTS

- **INTEGRITY: HANDLING ERRORS AND BAD DATA**
- **SECURITY: IMPLEMENT BCrypt FOR PASSWORD HASHING AND ENVIRONMENT VARIABLES FOR DB CREDENTIALS.**
- **USABILITY: UI MUST BE SIMPLE AND MOBILE-FRIENDLY**
- **RELIABILITY: SYSTEM SHOULD CONSISTENTLY BE UP**

5. SYSTEM ARCHITECTURE

- **REST ENDPOINTS: (LIST METHOD | URL | DESCRIPTION):**

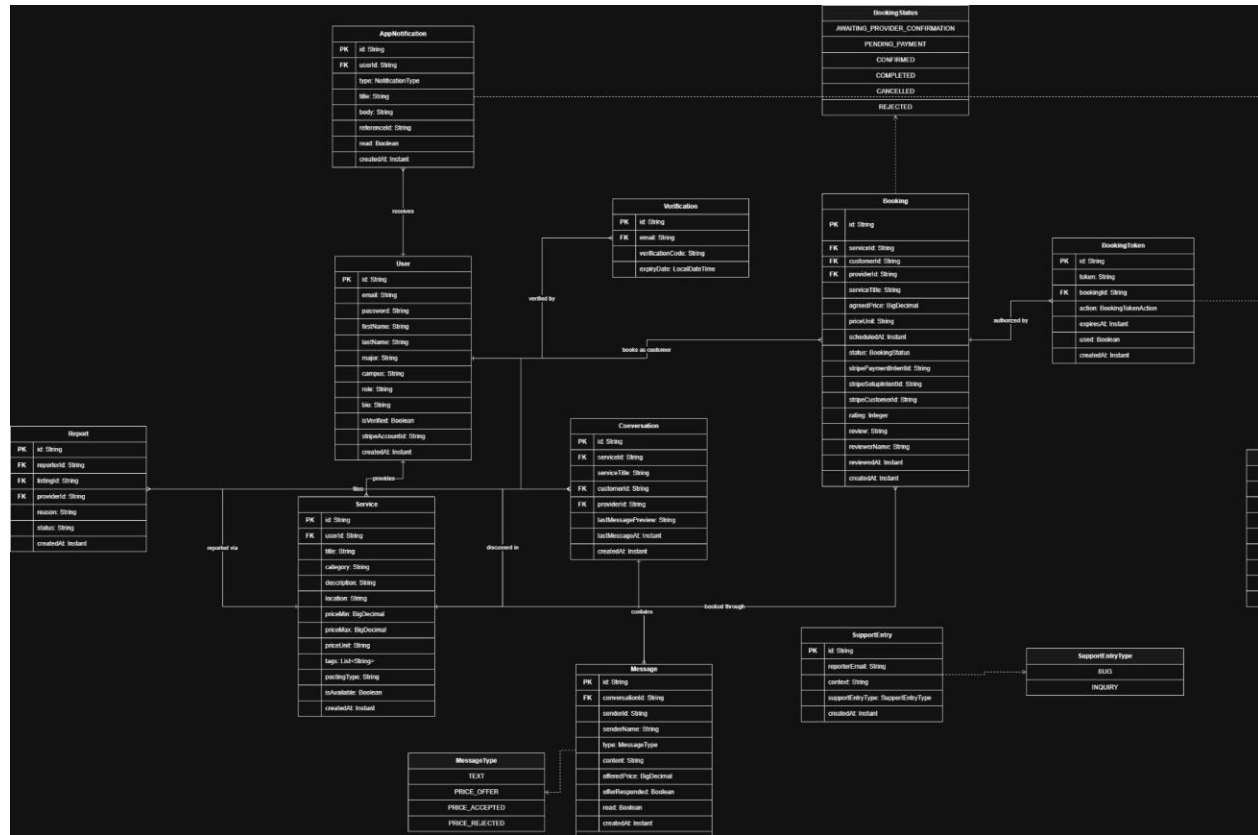
- **LIST YOUR PLANNED ENDPOINTS BASED ON YOUR REST SLIDE:**

- **GET /API/RESOURCES — FETCHES COLLECTION.**

- **POST /API/RESOURCES — CREATES A NEW RESOURCE.**

<u>METHOD</u>	<u>URL</u>	<u>DESCRIPTION</u>
○ POST	/api/auth/register	Creates a new user account.
○ POST	/api/auth/login	Logs in a user.
○ POST	/api/auth/logout	Logs out the current user.
○ GET	/api/services	Fetches all service listings.
○ POST	/api/services	Creates a new service listing.
○ PUT	/api/services/{serviceId}	Updates a service listing.
○ DELETE	/api/services/{serviceId}	Deletes a service listing.
○ GET	/api/users/me	Fetches the current user profile.
○ PATCH	/api/users/me	Updates the current user profile.
○ POST	/api/bookings	Creates a new booking request.
○ GET	/api/bookings/customer/me	Fetches a customer's bookings.
○ GET	/api/bookings/provider/requests/me	Fetches provider booking requests.
○ PUT	/api/bookings/{id}/confirm	Confirms a booking.
○ PUT	/api/bookings/{id}/cancel	Cancels a booking.
○ DELETE	/api/bookings/{id}/reject	Rejects a booking.
○ POST	/api/conversations	Starts a conversation.
○ GET	/api/conversations	Fetches user conversations.
○ POST	/api/conversations/{conversationId}/messages	Sends a message.
○ GET	/api/notifications	Fetches user notifications.
○ POST	/api/reports	Creates a listing or provider report.
○ POST	/api/support/bug	Submits a bug report.
○ POST	/api/verification/code	Verifies an email code.

- **UML: (ATTACH A DIAGRAM)**



5. USER INTERFACE (UI)

A. 5.1 WIREFRAMES / MOCKUPS

6. DATA REQUIREMENTS:

A.

ID	REQUIREMENT	PRIORITY
DR-01	The system shall store user account and profile data, including email, password, name, major, campus, role, bio, verification status, Stripe account, ID, and creation date	1
DR-02	The system shall store service listing data, including title, category,	2

	provider ID, price range, description, location, tags, posting type, availability, and creation date	
DR-03	The system shall store booking data, including service ID, customer ID, provider ID, agreed price, scheduled time, booking status, payment IDs, review, rating, and creation date	3
DR-04	The system shall store conversation and message data between customers and providers	4
DR-05	The system shall store notification data, including receipt, notification type, title, message body, reference ID, read status, and creation date	5
DR-06	The system shall store verification data for account confirmation, including email, verification code, and expiration date.	6
DR-07	The system shall store support request data, including reporter email,	7

	request context, support type, and creation date	
DR-08	The system shall store report data for flagged listings or providers, including reporter ID, listing ID, provider ID, reason, status, and creation date.	8
DR-09	The system shall store booking action token data, including tokens, booking ID, action type, expiration date, used status, and creation date	9

7. **TRACEABILITY MATRIX**

ID	REQUIREMENT	LINE OF CODE
DR-01	STORE USER ACCOUNT AND PROFILE DATA	USER.JAVA - LINE 16
DR-02	STORE SERVICE LISTING DATA	SERVICE.JAVA - LINE 17
DR-03	STORE BOOKING DATA	BOOKING.JAVA- LINE 14
DR-04	STORE CONVERSATION AND MANAGE DATA	CONVERSATION.JAVA - LINE 13, MESSAGE.JAVA-LINE 14
DR-05	STORE NOTIFICATION DATA	APPNOTIFICATION.JAVA - LINE 13
DR-06	STORE VERIFICATION	VERIFICATION.JAVA -

	DATA	LINE 13
DR-07	STORE SUPPORT REQUEST DATA	SUPPORTENTRY.JAVA - LINE 15
DR-08	STORE REPORT DATA	REPORT.JAVA - LINE 12
DR-09	STORE BOOKING ACTION TOKEN DATA	BOOKINGTOKEN.JAVA - LINE 13

8. **AI USAGE & DISCLOSURE (MANDATORY)**

- **MODEL(S) USED:** [E.G., CLAUDE 3.5, GPT-4O]
- **PROMPTS USED DURING CODING:**

PLANNING AND ARCHITECTURE:

- "WHAT'S THE BEST APPROACH FOR BUILDING X FEATURE?"
- "WHAT FILES WILL BE AFFECTED BY THIS CHANGE?"
- "SHOULD WE PUT THIS IN ONE BRANCH OR TWO?"

IMPLEMENTATION PROMPTS TO CODEX:

- "CREATE A NEW COMPONENT X THAT DOES Y, STYLED LIKE Z"
- "ADD ENDPOINT X TO CONTROLLER Y FOLLOWING THE SAME PATTERN AS Z"
- "REFACTOR X TO MOVE LOGIC FROM CONTROLLER TO SERVICE LAYER"
- "FIX CONFLICT MARKERS IN FILE X"
- "ADD FIELD X TO DTO Y AND POPULATE IT IN SERVICE Z"

BUG FIXING:

- "THE SUSPEND BUTTON IS STILL SPAMMABLE, KEEP IT DISABLED FOR 1 SECOND AFTER THE ACTION"
- "FIX CSS IMPORT PATH CASE SENSITIVITY IN MAJORCOMBOBOX"
- "THE SIDEBAR IS SQUISHING ON THE ADMIN PAGE, FIX THE LAYOUT"

REVIEW ASSISTANCE:

- "REVIEW THIS FILE FOR BUGS, LOGIC ERRORS, AND BEST PRACTICES"
- "WRITE A GITHUB REVIEW COMMENT FOR THIS ISSUE"
- "DO MY COMMENTS OVERLAP WITH SEAN'S?"